

Rensselaer Polytechnic Institute
Department of Electrical, Computer, and Systems Engineering
ECSE 2500: Engineering Probability, Spring 2026

Homework #1: due Thursday, Jan. 22nd, at 2PM.

Show all work for full credit!

Submit your work as a single PDF on Gradescope, labeling each problem number with a page.

In this homework, we will investigate the whereabouts of 5 kids: Max, Eleven, Will, Holly, and Derek. Each kid's body can be in one of three physical locations: Hawkins, the Upside Down, or the Abyss. Each kid's mind can be in one of two places: inside their physical body or inside Vecna's mind.

1. For each of the below experiments, determine the sample space. If the sample space is finite, determine the number of outcomes. If the number of outcomes is too large, you don't have to write them all down explicitly, just give a few examples to indicate the pattern.
 - (a) (5 points.) We record the number of times that Will has visited the Upside Down.
 - (b) (5 points.) We record the number of times Eleven has been in Vecna's mind minus the number of times Max has been in Vecna's mind.
 - (c) (5 points.) At a given moment, we record the ordered pair (number of kids' minds in their own bodies, number of kids' minds in Vecna's mind).
 - (d) (5 points.) At a given moment, we record the ordered pair (number of kids in Hawkins, number of kids in the Upside Down, number of kids in the Abyss).
 - (e) (5 points.) At a given moment, we record the unordered set of the kids currently in Hawkins.
 - (f) (5 points.) We record the total amount of time Holly has spent in the Abyss.
 - (g) (5 points.) We record the order in which the kids first arrive to the Upside Down (assuming that they all do at some point).
 - (h) (10 points.) At a given moment, we record the (body, mind) locations of each of the five kids in the order (Max, Eleven, Will, Holly, Derek).
2. In this question we consider the setup of the last part of Problem 1, and assume that each of these outcomes is equally probable. Compute the probability of the following events.
 - (a) (10 points.) No kids are mentally in their own bodies.
 - (b) (10 points.) At least 2 kids' (body, mind) locations match.
 - (c) (10 points.) At least 3 kids' bodies are in the Upside Down.
 - (d) (10 points.) The girls (Max, Eleven, and Holly) are all in the same physical location and the boys (Will and Derek) are both in the same mental location.
3. (15 points.) In the Abyss, Vecna has a Pain Tree with 12 slots arranged in a line where kids can be plugged in. Suppose that all 5 kids arrive at the Abyss eventually, and as soon a kid arrives, Vecna plugs them into the pain tree, selecting a slot at random from the remaining empty slots. Compute the probability that after all the kids have arrived, no two kids are in adjacent slots. (This is a bit tricky, and is related to a combinatorial technique called "stars and bars".)